



CORE CS (THEORY)

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❖ Top 50 DBMS Interview Questions

1. What is DBMS?

DBMS stands for Database Management System. It is software that helps store, retrieve, and manage data efficiently. DBMS provides security, consistency, and data integrity. Examples include MySQL, Oracle, and SQL Server.

2. What is a Database?

A database is an organized collection of data stored electronically. It allows efficient access, management, and updating of data. Databases reduce data redundancy. They are used in almost every application today.

3. What are the advantages of DBMS?

DBMS reduces data redundancy and inconsistency. It provides data security and integrity. It supports concurrent access and backup recovery. DBMS improves data sharing and reliability.

4. What is Data Redundancy?

Data redundancy means duplication of data in multiple places. It increases storage cost and inconsistency. DBMS minimizes redundancy using normalization. Less redundancy improves data accuracy.

5. What is Data Integrity?

Data integrity ensures accuracy and consistency of data. It is maintained using constraints like primary key and foreign key. Integrity prevents invalid data entry. It is essential for reliable databases.

6. What is Data Independence?

Data independence means changes in data structure do not affect application programs. It allows modifying schema without rewriting code. There are logical and physical data independence. It improves system flexibility.

7. What is Schema?

Schema defines the structure of a database. It includes tables, fields, and relationships. Schema is defined during database design. It acts as a blueprint of the database.

8. What is Instance?

An instance is the actual data stored in the database at a particular time. It changes frequently with operations. Instance reflects the current state of the database. Schema remains constant compared to instance.

9. What is a Table?

A table is a collection of related data organized in rows and columns. Each row represents a record. Each column represents an attribute. Tables are basic storage units in DBMS.

10. What is a Primary Key?

A primary key uniquely identifies each record in a table. It cannot contain null values. Each table can have only one primary key. It ensures entity integrity.

11. What is a Foreign Key?

A foreign key is a field that refers to the primary key of another table. It establishes relationships between tables. It maintains referential integrity. It prevents invalid data entries.

12. What is a Candidate Key?

A candidate key is an attribute that can uniquely identify records. A table can have multiple candidate keys. One of them is chosen as the primary key. Others are called alternate keys.

13. What is a Composite Key?

A composite key consists of two or more attributes. Together they uniquely identify a record. It is used when a single attribute is not enough. Common in junction tables.

14. What is a Super Key?

A super key is a set of attributes that uniquely identifies records. It may contain extra attributes. All candidate keys are super keys. Super keys ensure uniqueness.

15. What is Normalization?

Normalization is the process of organizing data to reduce redundancy. It divides tables into smaller related tables. It improves data consistency. Normal forms define normalization rules.

16. What is Denormalization?

Denormalization is the opposite of normalization. It combines tables to improve performance. It increases redundancy intentionally. Used in read-heavy systems.

17. What is 1NF?

First Normal Form ensures atomic values in columns. Each column contains single values only. It removes repeating groups. It is the basic level of normalization.

18. What is 2NF?

Second Normal Form removes partial dependency. It requires 1NF compliance. Non-key attributes depend on the full primary key. Used mainly with composite keys.

19. What is 3NF?

Third Normal Form removes transitive dependency. It requires 2NF compliance. Non-key attributes depend only on primary key. It improves data integrity.

20. What is BCNF?

BCNF is a stronger form of 3NF. Every determinant must be a candidate key. It removes anomalies not handled by 3NF. Used in advanced normalization.

21. What is SQL?

SQL stands for Structured Query Language. It is used to interact with databases. SQL allows data retrieval, insertion, update, and deletion. It is standardized and widely used.

22. What are DDL Commands?

DDL stands for Data Definition Language. It defines database structure. Commands include CREATE, ALTER, DROP, and TRUNCATE. DDL changes schema.

23. What are DML Commands?

DML stands for Data Manipulation Language. It is used to manage data. Commands include INSERT, UPDATE, and DELETE. DML works on table data.

24. What are DCL Commands?

DCL stands for Data Control Language. It controls database access. Commands include GRANT and REVOKE. DCL ensures security.

25. What are TCL Commands?

TCL stands for Transaction Control Language. It manages transactions. Commands include COMMIT, ROLLBACK, and SAVEPOINT. TCL ensures consistency.

26. What is a Transaction?

A transaction is a sequence of operations performed as a single unit. It ensures data consistency. Transactions follow ACID properties. Either all operations succeed or fail.

27. What is ACID Property?

ACID stands for Atomicity, Consistency, Isolation, and Durability. These properties ensure reliable transactions. They maintain data integrity. Essential for transaction processing.

28. What is Atomicity?

Atomicity ensures a transaction is fully completed or not at all. Partial updates are not allowed. It prevents inconsistent states. Maintained using rollback.

29. What is Consistency?

Consistency ensures database moves from one valid state to another. It maintains integrity constraints. Transactions must follow rules. Invalid data is not allowed.

30. What is Isolation?

Isolation ensures transactions execute independently. Concurrent transactions do not affect each other. Prevents anomalies like dirty reads. Achieved using locks.

31. What is Durability?

Durability ensures committed data is permanent. Data remains safe even after failure. Maintained using logs and backups. Ensures reliability.

32. What is a View?

A view is a virtual table created using SQL query. It does not store data physically. Views improve security and simplicity. Used for abstraction.

33. What is an Index?

An index improves data retrieval speed. It works like a book index. Indexes reduce search time. They increase storage usage.

34. What is a Clustered Index?

A clustered index sorts table data physically. Only one clustered index is allowed per table. It improves range queries. Data order is changed.

35. What is a Non-Clustered Index?

Non-clustered index does not change physical order. It stores pointers to data. Multiple non-clustered indexes are allowed. Improves lookup performance.

36. What is a Cursor?

A cursor is a database object used to retrieve records one by one. It is useful for row-level operations. Cursors increase flexibility. Mostly used in stored procedures.

37. What is a Stored Procedure?

A stored procedure is a set of SQL statements saved in database. It improves performance and security. Procedures can accept parameters. They reduce network traffic.

38. What is a Trigger?

A trigger is automatically executed in response to database events. It fires on INSERT, UPDATE, or DELETE. Triggers enforce business rules. Used for auditing.

39. What is a Join?

A join combines records from two or more tables. It is based on related columns. Types include inner, left, right, and full join. Joins retrieve meaningful data.

40. What is Inner Join?

Inner join returns matching records from both tables. Non-matching rows are excluded. It is the most commonly used join. Improves data accuracy.

41. What is Left Join?

Left join returns all records from left table. Matching records from right table are included. Non-matching rows return null. Useful for optional relationships.

42. What is Right Join?

Right join returns all records from right table. Matching records from left table are included. Non-matching rows return null. Opposite of left join.

43. What is Full Join?

Full join returns records from both tables. Includes matching and non-matching rows. Non-matches return null values. Useful for complete data comparison.

44. What is Subquery?

A subquery is a query inside another query. It is used for complex data retrieval. Subqueries can return single or multiple values. Improves query flexibility.

45. What is Constraint?

A constraint enforces rules on table data. It ensures data accuracy and integrity. Types include NOT NULL, UNIQUE, CHECK, and DEFAULT. Constraints prevent invalid data.

46. What is NOT NULL Constraint?

NOT NULL constraint prevents null values in a column. It ensures mandatory data entry. Improves data completeness. Used for important fields.

47. What is UNIQUE Constraint?

UNIQUE constraint ensures unique values in a column. It allows one null value. Prevents duplicate data. Useful for email or username fields.

48. What is CHECK Constraint?

CHECK constraint limits values in a column. It ensures data validity. Example: age > 18. Used for validation rules.

49. What is DEFAULT Constraint?

DEFAULT constraint assigns default value when no value is provided. It simplifies data insertion. Improves consistency. Example: default status = active.

50. Why use DBMS?

DBMS provides efficient data storage and retrieval. It ensures security and integrity. Supports multi-user access. Essential for modern applications.

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